SIX EXPLICIT WORDS IN THE ASKEY-WILSON GENERATORS DEFINE ONE AND THE SAME ELEMENT

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Allowed standard prerequisites (editorial).

- The defining relations of the universal Askey-Wilson algebra, arranged as the relations involving alpha, beta, gamma. Same source, Definition labelled def:abc; invoked inside the counted proof at lines 1511-1512 and 1526-1527. Supplies the two relations whose vanishing makes the displayed differences zero.
- The automorphism rho of the universal Askey-Wilson algebra. Same source, set up before the statement; used inside the counted proof at lines 1495-1501. Cyclically permutes the six displayed elements, which reduces the verification to two computations.

Source reference. Original definition 1.3: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1512–1512. Navigation pointer for \ref{def:abc} used inside the counted span (line 1512); the argument that consumes it is retained in the span.

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Lemma 0.1. *The following elements of* Δ *coincide:*

$$qABC + q^2A^2 + q^{-2}B^2 + q^2C^2 - qA\alpha - q^{-1}B\beta - qC\gamma,$$
$$qBCA + q^2A^2 + q^2B^2 + q^{-2}C^2 - qA\alpha - qB\beta - q^{-1}C\gamma,$$
$$qCAB + q^{-2}A^2 + q^2B^2 + q^2C^2 - q^{-1}A\alpha - qB\beta - qC\gamma,$$
$$q^{-1}CBA + q^{-2}A^2 + q^2B^2 + q^{-2}C^2 - q^{-1}A\alpha - qB\beta - q^{-1}C\gamma,$$
$$q^{-1}ACB + q^{-2}A^2 + q^{-2}B^2 + q^2C^2 - q^{-1}A\alpha - q^{-1}B\beta - qC\gamma,$$
$$q^{-1}BAC + q^2A^2 + q^{-2}B^2 + q^{-2}C^2 - qA\alpha - q^{-1}B\beta - q^{-1}C\gamma.$$

We denote this common element by Ω.

Proof. Denote the displayed sequence of elements by Ω_B^+, Ω_C^+, Ω_A^+, Ω_B^-, Ω_C^-, Ω_A^-. The automorphism ρ cyclically permutes Ω_A^+, Ω_B^+, Ω_C^+ and cyclically permutes Ω_A^-, Ω_B^-, Ω_C^-. The element $\Omega_B^+ - \Omega_C^-$ is equal to $(q - q^{-1})A$ times

$$\big(q + q^{-1}\big)A + \frac{qBC - q^{-1}CB}{q - q^{-1}} - \alpha. \tag{1}$$

The element (1) is zero by Definition 1.3 so $\Omega_B^+ = \Omega_C^-$. In this equation we apply ρ twice to get $\Omega_C^+ = \Omega_A^-$ and $\Omega_A^+ = \Omega_B^-$. The element $\Omega_B^+ - \Omega_A^-$ is equal to

$$\big(q + q^{-1}\big)C + \frac{qAB - q^{-1}BA}{q - q^{-1}} - \gamma \tag{2}$$

times $(q - q^{-1})C$. The element (2) is zero by Definition 1.3 so $\Omega_B^+ = \Omega_A^-$. Applying ρ twice we get $\Omega_C^+ = \Omega_B^-$ and $\Omega_A^+ = \Omega_C^-$. By these comments Ω_B^+, Ω_C^+, Ω_A^+, Ω_B^-, Ω_C^-, Ω_A^- coincide. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.